Math 151A Homework 7
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Problem 1: Section 6.1, Exercise 3

Use Gaussian elimination with backward substitution and two digit rounding arithmetic. The exact solution is $x_1 = 1$, $x_2 = -1$, $x_3 = 3$.

(a)

$$\begin{cases} 4x_1 - x_2 + x_3 = 8, \\ 2x_1 + 5x_2 + 2x_3 = 3, \\ x_1 + 2x_2 + 4x_3 = 11. \end{cases}$$

Start with the augmented matrix

$$\left[\begin{array}{ccc|c} 4 & -1 & 1 & 8 \\ 2 & 5 & 2 & 3 \\ 1 & 2 & 4 & 11 \end{array}\right].$$

Elimination in column 1.

$$m_{21} = \frac{2}{4} = 0.5, \qquad m_{31} = \frac{1}{4} = 0.25.$$

Row operations (each arithmetic rounded to two digits):

$$\begin{aligned} R_2 \leftarrow R_2 - 0.5R_1 : \quad & [2, 5, 2 \,|\, 3] - 0.5[4, -1, 1 \,|\, 8] \\ & = [0, 5.5, 1.5 \,|\, -1], \\ R_3 \leftarrow R_3 - 0.25R_1 : \quad & [1, 2, 4 \,|\, 11] - 0.25[4, -1, 1 \,|\, 8] \\ & = [0, 2.2, 3.8 \,|\, 9]. \end{aligned}$$

Now the matrix is

$$\left[\begin{array}{ccc|c} 4 & -1 & 1 & 8 \\ 0 & 5.5 & 1.5 & -1 \\ 0 & 2.2 & 3.8 & 9 \end{array}\right].$$

Elimination in column 2.

$$m_{32} = \frac{2.2}{5.5} \approx 0.40.$$

$$\begin{aligned} R_3 &\leftarrow R_3 - 0.40R_2 : \\ [0, 2.2, 3.8 \,|\, 9] &- 0.40[0, 5.5, 1.5 \,|\, -1] \\ &= [0, 0, 3.2 \,|\, 9.4]. \end{aligned}$$

Upper triangular system:

$$\left[\begin{array}{ccc|c} 4 & -1 & 1 & 8 \\ 0 & 5.5 & 1.5 & -1 \\ 0 & 0 & 3.2 & 9.4 \end{array}\right].$$

Backward substitution.

$$x_3 = \frac{9.4}{3.2} \approx 2.9.$$

$$5.5x_2 + 1.5x_3 = -1 \Rightarrow 5.5x_2 + 1.5(2.9) = -1.$$

$$1.5(2.9) \approx 4.4, \quad 5.5x_2 = -1 - 4.4 = -5.4, \quad x_2 \approx \frac{-5.4}{5.5} \approx -0.98.$$

$$4x_1 - x_2 + x_3 = 8 \Rightarrow 4x_1 - (-0.98) + 2.9 = 8.$$

$$-(-0.98) + 2.9 \approx 3.9, \quad 4x_1 = 8 - 3.9 = 4.1, \quad x_1 \approx \frac{4.1}{4} \approx 1.0.$$

So with two digit rounding,

$$\boxed{x_1 \approx 1.0,\ x_2 \approx -0.98,\ x_3 \approx 2.9}.$$

(b)

$$\begin{cases} 4x_1 + x_2 + 2x_3 = 9, \\ 2x_1 + 4x_2 - x_3 = -5, \\ x_1 + x_2 - 3x_3 = -9. \end{cases}$$

Augmented matrix

$$\left[\begin{array}{ccc|c} 4 & 1 & 2 & 9 \\ 2 & 4 & -1 & -5 \\ 1 & 1 & -3 & -9 \end{array}\right].$$

Elimination in column 1.

$$m_{21} = \frac{2}{4} = 0.5, \qquad m_{31} = \frac{1}{4} = 0.25.$$

$$\begin{aligned} R_2 \leftarrow R_2 - 0.5R_1 : \quad & [2, 4, -1 \,|\, -5] - 0.5[4, 1, 2 \,|\, 9] \\ & = [0, 3.5, -2 \,|\, -9.5], \\ R_3 \leftarrow R_3 - 0.25R_1 : \quad & [1, 1, -3 \,|\, -9] - 0.25[4, 1, 2 \,|\, 9] \\ & = [0, 0.75, -3.5 \,|\, -11]. \end{aligned}$$

Matrix:

$$\left[\begin{array}{ccc|c} 4 & 1 & 2 & 9 \\ 0 & 3.5 & -2 & -9.5 \\ 0 & 0.75 & -3.5 & -11 \end{array}\right].$$

Elimination in column 2.

$$m_{32} = \frac{0.75}{3.5} \approx 0.21.$$

$$\begin{aligned} R_3 \leftarrow R_3 - 0.21R_2 : \\ [0, 0.75, -3.5 \,|\, -11] - 0.21[0, 3.5, -2 \,|\, -9.5] \\ = [0, 0.01, -3.1 \,|\, -9.0]. \end{aligned}$$

Upper triangular system:

$$\left[\begin{array}{ccc|c} 4 & 1 & 2 & 9 \\ 0 & 3.5 & -2 & -9.5 \\ 0 & 0.01 & -3.1 & -9.0 \end{array}\right].$$

Backward substitution.

$$x_3 = \frac{-9.0}{-3.1} \approx 2.9.$$

$$3.5x_2 - 2x_3 = -9.5 \Rightarrow 3.5x_2 - 2(2.9) = -9.5.$$

$$2(2.9) \approx 5.8, \quad 3.5x_2 = -9.5 + 5.8 = -3.7, \quad x_2 \approx \frac{-3.7}{3.5} \approx -1.1.$$

$$4x_1 + x_2 + 2x_3 = 9 \Rightarrow 4x_1 - 1.1 + 2(2.9) = 9.$$

$$2(2.9) \approx 5.8, \quad -1.1 + 5.8 = 4.7, \quad 4x_1 = 9 - 4.7 = 4.3, \quad x_1 \approx \frac{4.3}{4} \approx 1.1.$$

So

$$\boxed{x_1 \approx 1.1,\ x_2 \approx -1.1,\ x_3 \approx 2.9}.$$

Problem 2: Section 6.2, Exercise 1

Find the row interchanges that are required to solve each system using Algorithm 6.1.

(a)

$$\begin{cases} x_1 - 5x_2 + x_3 = 7, \\ 10x_1 + 20x_3 = 6, \\ 5x_1 - x_3 = 4. \end{cases}$$

Running Algorithm 6.1, the pivots in positions $(1,1)$, $(2,2)$ and $(3,3)$ are all nonzero after elimination, so no row interchange is needed.

$$\boxed{\text{No row interchanges are required.}}$$

(b)

$$\begin{cases} x_1 + x_2 - x_3 = 1, \\ x_1 + x_2 + 4x_3 = 2, \\ 2x_1 - x_2 + 2x_3 = 3. \end{cases}$$

In column 1 the pivot $a_{11} = 1$ is nonzero. After eliminating below, the entry in position $(2,2)$ becomes zero while the entry $(3,2)$ is nonzero, so Algorithm 6.1 must interchange row 2 and row 3 at this stage.

$$\boxed{\text{Interchange rows 2 and 3.}}$$

(c)

$$\begin{cases} 2x_1 - 3x_2 + 2x_3 = 5, \\ -4x_1 + 2x_2 - 6x_3 = 14, \\ 2x_1 + 2x_2 + 4x_3 = 8. \end{cases}$$

Here the pivots in positions $(1,1)$, $(2,2)$ and $(3,3)$ are all nonzero during the elimination, so the algorithm proceeds with no swaps.

$$\boxed{\text{No row interchanges are required.}}$$

(d)

$$\begin{cases} x_2 + x_3 = 6, \\ x_1 - 2x_2 - x_3 = 4, \\ x_1 - x_2 + x_3 = 5. \end{cases}$$

The first equation has coefficient 0 in front of x_1, so $a_{11} = 0$ while there are nonzero entries in that column below. Algorithm 6.1 must swap row 1 and row 2 to get a nonzero pivot in position $(1,1)$.

$$\boxed{\text{Interchange rows 1 and 2.}}$$

Problem 3: Section 7.1, Exercise 7

For each system $Ax = b$ with exact solution x and approximate solution $\tilde{x}$, compute

$$\|x - \tilde{x}\|_\infty \quad \text{and} \quad \|A\tilde{x} - b\|_\infty.$$

(a)

$$\frac{1}{2}x_1 + \frac{1}{3}x_2 = \frac{1}{63},$$
$$\frac{1}{3}x_1 + \frac{1}{4}x_2 = \frac{1}{168},$$
$$x = \left(\frac{1}{7},\ -\frac{1}{6}\right)^T, \quad \tilde{x} = (0.142,\ -0.166)^T.$$

Error vector:

$$x - \tilde{x} = \left(\frac{1}{7} - 0.142,\ -\frac{1}{6} + 0.166\right) \approx (0.00086,\ -0.00067).$$

$$\|x - \tilde{x}\|_\infty \approx 0.00086.$$

Residual:

$$A\tilde{x} - b = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 0.142 \\ -0.166 \end{pmatrix} - \begin{pmatrix} \frac{1}{63} \\ \frac{1}{168} \end{pmatrix} \approx \begin{pmatrix} -0.00021 \\ -0.00012 \end{pmatrix}.$$

$$\|A\tilde{x} - b\|_\infty \approx 0.00021.$$

(b)

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 1, \\ 2x_1 + 3x_2 + 4x_3 &= -1, \\ 3x_1 + 4x_2 + 6x_3 &= 2, \end{aligned}$$

$$x = (0,\ -7,\ 5)^T, \quad \tilde{x} = (-0.33,\ -7.9,\ 5.8)^T.$$

Error:

$$x - \tilde{x} = (0.33,\ 0.9,\ -0.8)^T, \quad \|x - \tilde{x}\|_\infty = 0.9.$$

Residual:

$$A\tilde{x} - b \approx (0.27,\ -0.16,\ 0.21)^T, \quad \|A\tilde{x} - b\|_\infty \approx 0.27.$$

(c)

Same A and b as part (b) but

$$\tilde{x} = (-0.2,\ -7.5,\ 5.4)^T.$$

Error:

$$x - \tilde{x} = (0.2,\ 0.5,\ -0.4)^T, \quad \|x - \tilde{x}\|_\infty = 0.5.$$

Residual:

$$A\tilde{x} - b \approx (0,\ -0.30,\ -0.20)^T, \quad \|A\tilde{x} - b\|_\infty \approx 0.30.$$

(d)

$$\begin{aligned}
0.04x_1 + 0.01x_2 - 0.01x_3 &= 0.06,\\
0.2x_1 + 0.5x_2 - 0.2x_3 &= 0.3,\\
x_1 + 2x_2 + 4x_3 &= 11,\\
x &= (1.827586,\ 0.6551724,\ 1.965517)^T,\\
\tilde{x} &= (1.8,\ 0.64,\ 1.9)^T.
\end{aligned}$$

Error:

$$x - \tilde{x} \approx (0.0276,\ 0.0152,\ 0.0655)^T, \quad \|x - \tilde{x}\|_\infty \approx 0.0655.$$

Residual:

$$A\tilde{x} - b \approx (-0.0006,\ 0,\ -0.32)^T, \quad \|A\tilde{x} - b\|_\infty \approx 0.32.$$

Problem 4: Section 7.3, Exercise 1

Find the first two Jacobi iterations for each system using $x^{(0)} = 0$.

(a)

$$\begin{cases} 3x_1 - x_2 + x_3 = 1,\\ 3x_1 + 6x_2 + 2x_3 = 0,\\ 3x_1 + 3x_2 + 7x_3 = 4. \end{cases}$$

Solve each equation for the corresponding variable:

$$\begin{aligned}
x_1 &= \frac{1 + x_2 - x_3}{3},\\
x_2 &= \frac{-3x_1 - 2x_3}{6},\\
x_3 &= \frac{4 - 3x_1 - 3x_2}{7}.
\end{aligned}$$

First iteration $x^{(1)}$ with $x^{(0)} = (0, 0, 0)^T$:

$$\begin{aligned}
x_1^{(1)} &= \frac{1 + 0 - 0}{3} = \frac{1}{3},\\
x_2^{(1)} &= \frac{-3 \cdot 0 - 2 \cdot 0}{6} = 0,\\
x_3^{(1)} &= \frac{4 - 3 \cdot 0 - 3 \cdot 0}{7} = \frac{4}{7}.
\end{aligned}$$

So

$$x^{(1)} = \left(\frac{1}{3},\ 0,\ \frac{4}{7}\right)^T.$$

Second iteration $x^{(2)}$:

$$x_1^{(2)} = \frac{1 + x_2^{(1)} - x_3^{(1)}}{3} = \frac{1 + 0 - \frac{4}{7}}{3} = \frac{3/7}{3} = \frac{1}{7},$$

$$x_2^{(2)} = \frac{-3x_1^{(1)} - 2x_3^{(1)}}{6} = \frac{-3 \cdot \frac{1}{3} - 2 \cdot \frac{4}{7}}{6} = \frac{-1 - \frac{8}{7}}{6} = \frac{-15/7}{6} = -\frac{5}{14},$$

$$x_3^{(2)} = \frac{4 - 3x_1^{(1)} - 3x_2^{(1)}}{7} = \frac{4 - 3 \cdot \frac{1}{3} - 0}{7} = \frac{3}{7}.$$

$$x^{(2)} = \left(\frac{1}{7},\ -\frac{5}{14},\ \frac{3}{7}\right)^T.$$

(b)

$$\begin{cases} 10x_1 - x_2 = 9, \\ -x_1 + 10x_2 - 2x_3 = 7, \\ -2x_2 + 10x_3 = 6. \end{cases}$$

Solve for each variable:

$$x_1 = \frac{9 + x_2}{10},$$
$$x_2 = \frac{7 + x_1 + 2x_3}{10},$$
$$x_3 = \frac{6 + 2x_2}{10}.$$

First iteration from $x^{(0)} = 0$:

$$x^{(1)} = (0.9,\ 0.7,\ 0.6)^T.$$

Second iteration:

$$x_1^{(2)} = \frac{9 + 0.7}{10} = 0.97,$$
$$x_2^{(2)} = \frac{7 + 0.9 + 2 \cdot 0.6}{10} = 0.91,$$
$$x_3^{(2)} = \frac{6 + 2 \cdot 0.7}{10} = 0.74.$$

$$x^{(2)} = (0.97,\ 0.91,\ 0.74)^T.$$

(c)

$$\begin{cases} 10x_1 + 5x_2 = 6, \\ 5x_1 + 10x_2 - 4x_3 = 25, \\ -4x_2 + 8x_3 - x_4 = -11, \\ -x_3 + 5x_4 = -11. \end{cases}$$

Solve:

$$\begin{aligned} x_1 &= \frac{6 - 5x_2}{10}, \\ x_2 &= \frac{25 - 5x_1 + 4x_3}{10}, \\ x_3 &= \frac{-11 + 4x_2 + x_4}{8}, \\ x_4 &= \frac{-11 + x_3}{5}. \end{aligned}$$

From $x^{(0)} = (0, 0, 0, 0)^T$ we get

$$x^{(1)} = (0.6,\ 2.5,\ -1.375,\ -2.2)^T.$$

Using $x^{(1)}$,

$$x^{(2)} \approx (-0.65,\ 1.65,\ -0.40,\ -2.475)^T.$$

(d)

$$\begin{cases} 4x_1 + x_2 + x_3 + x_5 = 6, \\ -x_1 - 3x_2 + x_3 + x_4 = 6, \\ 2x_1 + x_2 + 5x_3 - x_4 - x_5 = 6, \\ -x_1 - x_2 - x_3 + 4x_4 = 6, \\ 2x_2 - x_3 + x_4 + 4x_5 = 6. \end{cases}$$

Solve each equation:

$$x_1 = \frac{6 - x_2 - x_3 - x_5}{4},$$
$$x_2 = \frac{-6 - x_1 - x_3 - x_4}{3},$$
$$x_3 = \frac{6 - 2x_1 - x_2 + x_4 + x_5}{5},$$
$$x_4 = \frac{6 + x_1 + x_2 + x_3}{4},$$
$$x_5 = \frac{6 - 2x_2 + x_3 - x_4}{4}.$$

Starting from $x^{(0)} = 0$:

$$x^{(1)} = (1.5,\ -2.0,\ 1.2,\ 1.5,\ 1.5)^T.$$

Using $x^{(1)}$:

$$x^{(2)} \approx (1.325,\ -1.60,\ 1.60,\ 1.675,\ 2.425)^T.$$

Problem 5: Section 6.6, Exercise 24

Suppose A and B are strictly diagonally dominant $n \times n$ matrices. Which of the following must be strictly diagonally dominant?

$$\text{(a) } -A, \quad \text{(b) } A^T, \quad \text{(c) } A + B, \quad \text{(d) } A^2, \quad \text{(e) } A - B.$$

Recall: A is strictly diagonally dominant if for each i,

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|.$$

(a) $-A$

The diagonal entry of $-A$ is $-a_{ii}$ and off diagonal entries are $-a_{ij}$. Absolute values are unchanged, so for each row

$$|-a_{ii}| = |a_{ii}| > \sum_{j \neq i} |a_{ij}| = \sum_{j \neq i} |-a_{ij}|.$$

So $-A$ is strictly diagonally dominant.

$$\boxed{\text{(a) must be strictly diagonally dominant.}}$$

(b) A^T

Transposing does not change any entry values on each row, only their positions. Row i of A^T contains the same numbers as column i of A, but the diagonal entry is still a_{ii} and the set of off diagonal entries has the same absolute values as in row i of A. Therefore the same inequality holds.

$$\boxed{\text{(b) must be strictly diagonally dominant.}}$$

(c) $A + B$

This need not be strictly diagonally dominant. Example (size 2):

$$A = \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} -3 & 1 \\ 0 & 2 \end{pmatrix}.$$

For A: $|3| > |2|$, $|3| > |1|$, so A is strictly diagonally dominant. For B: $|2| > |0|$ and $|-3| > |1|$, so B is also strictly diagonally dominant.

But

$$A + B = \begin{pmatrix} 0 & 3 \\ 1 & 5 \end{pmatrix},$$

and in the first row $|0| \not> |3|$. So $A + B$ is not strictly diagonally dominant.

$$\boxed{\text{(c) does not have to be strictly diagonally dominant.}}$$

(d) A^2

Again this need not be strictly diagonally dominant. Take

$$A = \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix},$$

which is strictly diagonally dominant as above.

Then

$$A^2 = AA = \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 11 & 12 \\ 6 & 11 \end{pmatrix}.$$

In the first row,

$$|11| \not> |12|,$$

so A^2 is not strictly diagonally dominant.

$$\boxed{\text{(d) does not have to be strictly diagonally dominant.}}$$

(e) $A - B$

This also need not be strictly diagonally dominant. Using the same A and

$$B = \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix},$$

both A and B are strictly diagonally dominant.

But

$$A - B = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix},$$

and in the first row $|1| \not> |3|$. So $A - B$ is not strictly diagonally dominant.

$$\boxed{\text{(e) does not have to be strictly diagonally dominant.}}$$

So overall, only $-A$ and A^T are guaranteed to be strictly diagonally dominant.